

# Data Cleaning & Visualization Project Report

**Title:**Employee Attrition Data Cleaning and Visualization using Python

## Introduction

This project focuses on cleaning and analyzing employee attrition data. Organizations collect employee information such as age, department, salary, overtime, and years of service. Raw datasets often contain missing values, duplicates, and outliers, which can affect analysis. The goal of this project is to preprocess the data and generate visual insights to understand workforce trends.

## Objective

The objective of this project is to:

- Clean raw employee data
- Handle missing values
- Remove duplicate records
- Detect and remove outliers
- Visualize key patterns in employee attrition
- Present insights in a meaningful way

## Tools Used

- Python
- Google Colab
- Pandas
- Matplotlib

## Dataset Description

The dataset contains employee-related information including:

- Age
- Attrition
- BusinessTravel
- DailyRate
- Department
- Distance From Home
- Education
- Education Field
- Employee Number
- Gender
- JobRole
- MaritalStatus
- MonthlyIncome

- Over Time
- Total Working Years
- Years At Company
- Years In CurrentRole
- Years Since Last Promotion

This dataset helps study employee work patterns and attrition behavior.

## Methodology

### Data Collection

The employee dataset was collected as a CSV file and imported into the notebook environment.

### Data Cleaning Process

#### 1. Handling Missing Values

- Missing numeric values were identified.
- Replaced using column mean.

#### 2. Removing Duplicates

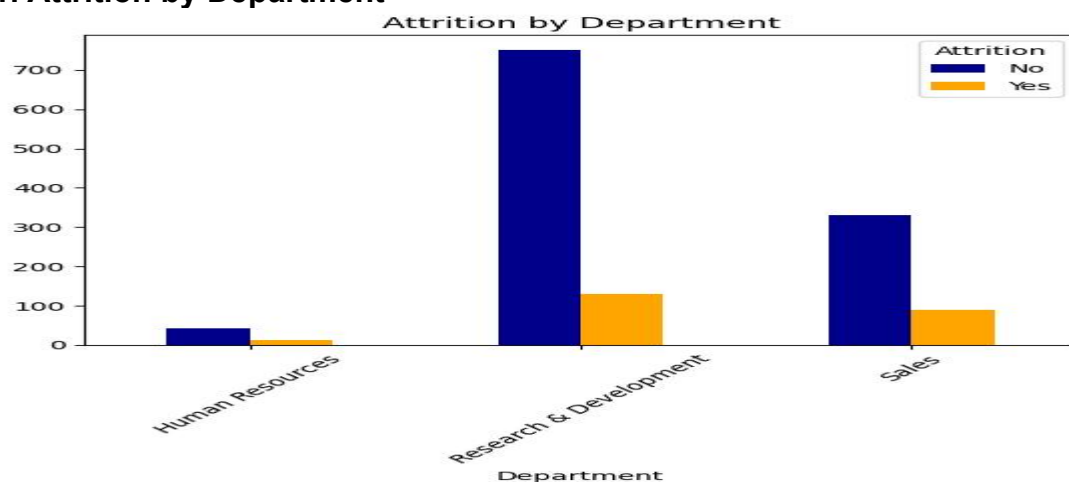
- Duplicate records were identified.
- Duplicate rows removed using Pandas.

#### 3. Outlier Detection

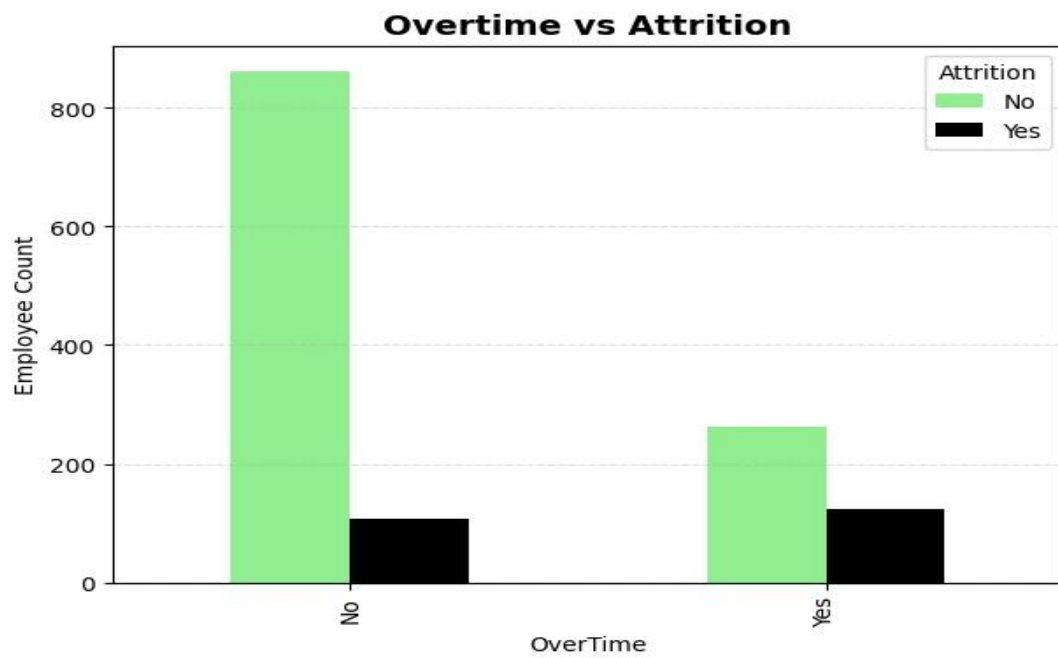
- Outliers detected in MonthlyIncome.
- IQR (Interquartile Range) method used.
- Extreme values removed.

## Visualizations Created

### 1. Attrition by Department

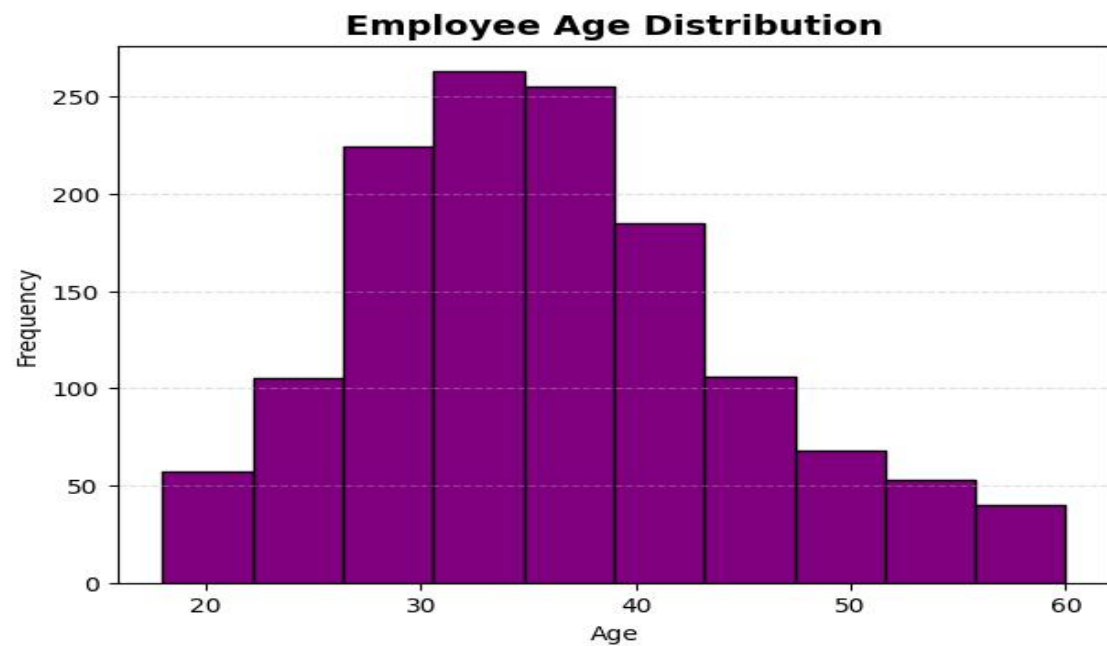


## 2. Overtime vs Attrition



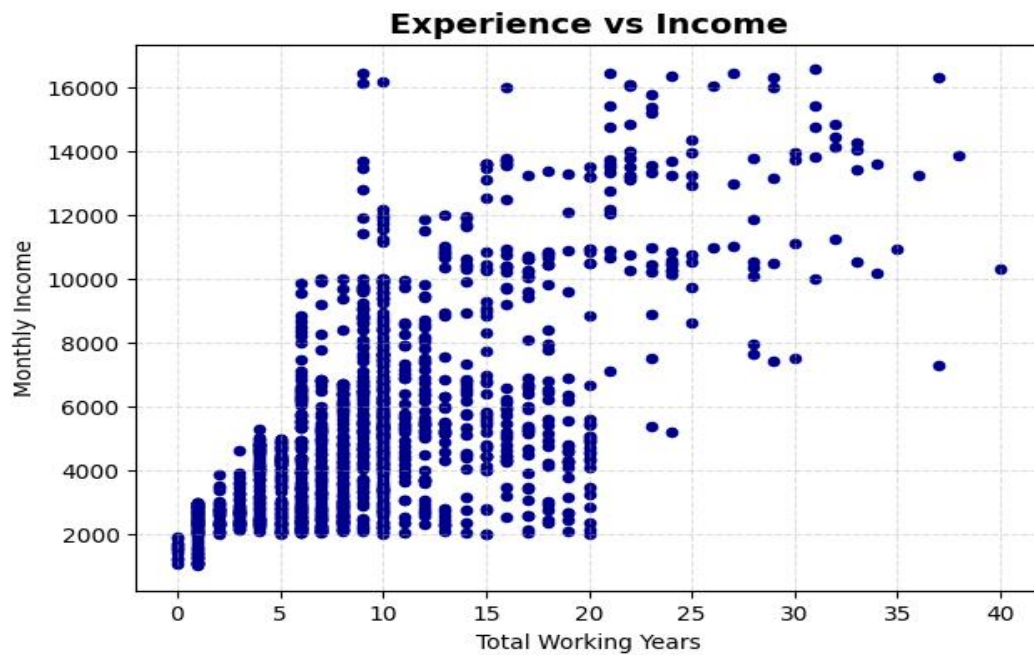
Shows how overtime impacts employee turnover.

## 3. Employee Age Distribution



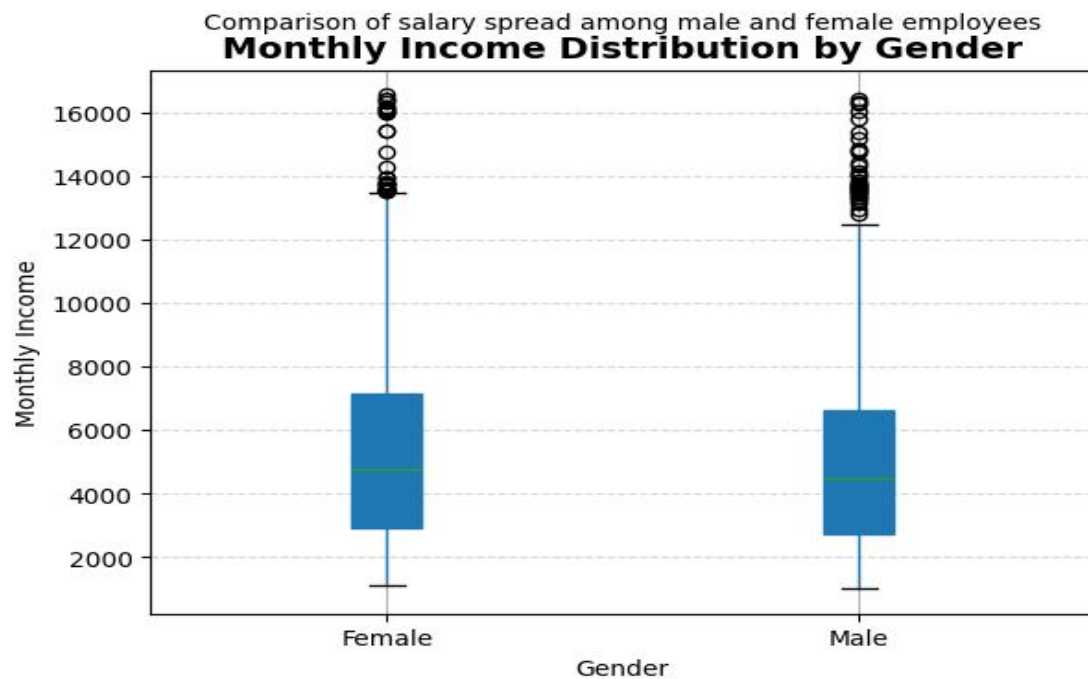
Histogram used to study employee age groups.

#### 4.Experience vs Income



Scatter plot used to show relationship between work experience and monthly income.

#### 5.Monthly Income Distribution by Gender



Boxplot used to compare salary distribution among genders.

## **Findings**

The analysis revealed the following:

- Employees who work overtime show higher attrition.
- Monthly income generally increases with total working years.
- Some departments have higher attrition than others.
- Most employees fall in the middle age group.
- Salary distribution varies across genders.
- Outlier salary values affected overall statistics and were removed.

## **Benefits of the Project**

This project improved understanding of:

- Data preprocessing
- Handling real-world noisy data
- Visualization techniques
- Data storytelling
- Business insight generation

## **Challenges Faced**

- Handling file upload issues
- Identifying correct columns for analysis
- Managing outliers in salary data
- Choosing meaningful visualizations

## **Conclusion**

The employee attrition dataset was successfully cleaned and analyzed. Missing values, duplicates, and outliers were handled effectively. Multiple charts were created to visualize attrition patterns, salary distribution, and workforce characteristics. The project demonstrates how raw HR data can be transformed into useful insights for organizational decision-making.

## **Future Scope**

Future enhancements can include:

- Building interactive dashboards in Microsoft Power BI
- Applying machine learning for attrition prediction
- Creating web dashboards
- Performing advanced statistical analysis